Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech. | *Semester- II* | *Time: 03 Hours*

Code: 100205 | *Subject:* Workshop Manufacturing Practices | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer any seven question of the following: [2 x 7 = 14]

(a) Define alloys.
(b) Differentiate between wood and timber.
(c) What are different types of allowances provided on pattern for casting?
(d) Differentiate between live centre and dead centre on lathe machine.
(e) Discuss the types of plastics.
(f) Write on the classification of steel.
(g) Which casting method is adopted for ornaments and toys of non-ferrous alloy?
(h) Explain recrystallization temperature.
(i) Why shear is provided on punches and dies in sheet metal blanking.
(j) Explain various forming operation.

Q.2 Explain the various techniques used for glass moulding. [14]

Q.3 (a) Draw and explain four types of seam in sheet metal operation. [7]
(b) List the various additive manufacturing process and explain 3 D printing in detail. [7]

Q.4 (a) List any six forging defects. [6]
(b) List out the forging operations. Explain any four forging operations with a suitable diagram. [8]

Q.5 (a) List any four materials used for pattern making. [2]
(b) Explain any four properties of molding sand. [8]
(c) What is meant by sand binder? Name the types of sand binders. [4]

Q.6 Explain the various operations that can be performed in fitting shop. [14]

Q.7 (a) Explain different types of flame in gas welding with suitable diagrams. [7]
(b) What do you understand by CNC machine? Explain with neat sketch the working of any CNC machine [7]

Q.8 Explain the different types of saw used in carpentry shop along with neat sketches. [14]

Q.9 Explain the types of welding joints. Also, clearly differentiate between soldering and brazing along with its applications. [14]

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